

Differential equations for the series of hypermaps with control on their full degree profile

Houcine Ben Dali

Université de Lorraine, CNRS, IECL, France

Université de Paris Cité, CNRS, IRIF, France



Seed seminar

May 29, 2024

Main result:

$$\begin{aligned} & \left(\mathcal{C}_\ell^{(\alpha)}(-t, \mathbf{q}) + \mathcal{C}_\ell^{(\alpha)}(-t, \mathbf{r}) \right) \cdot G^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) \\ &= \mathcal{C}_\ell^{(\alpha)\perp}(-t, \mathbf{p}) \cdot G^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}), \quad \text{for } \ell \geq 0; \end{aligned}$$

$G^{(\alpha)}$ is a deformation of the series of hypermaps with controlled degrees of vertices, faces and hyperedges.

- ▶ series of orientable hypermaps for $\alpha = 1$;
- ▶ series of all hypermaps (orientable or not) if $\alpha = 2$.

$\mathcal{C}_\ell^{(\alpha)}$ is a differential operator.

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Proof:

- Algebraic for any α : Jack characters.
- Combinatorial proof for $\alpha = 1$: orientable maps.

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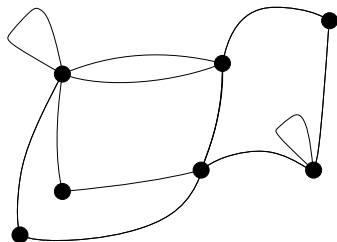
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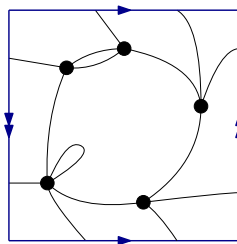
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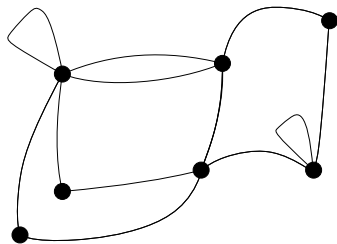
A planar map (a map on the sphere)



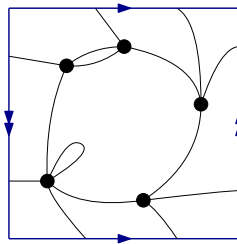
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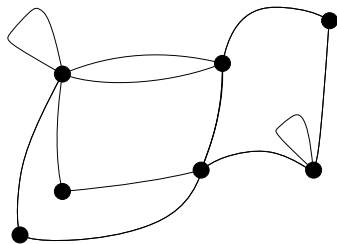
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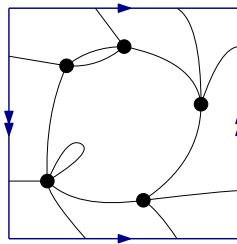
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- A *connected map* is a cellular embedding of a connected graph in a surface, **orientable or not**, considered up to homeomorphism.
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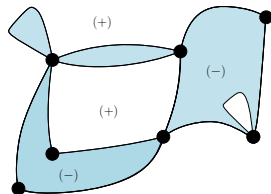
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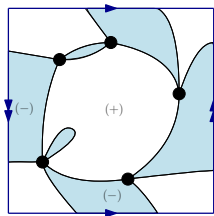
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- A hypermap is a map whose faces are colored in two colors (+) and (-), and such that each edge separates two faces of different colors.



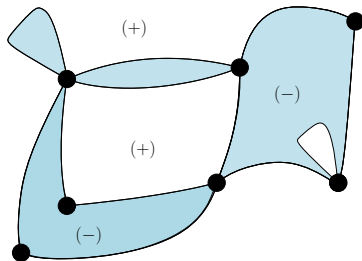
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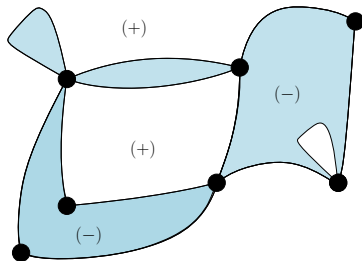
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Hypermaps

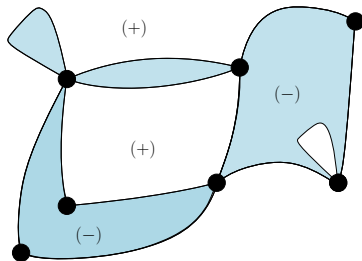
- The **size** of a map M is its number of edges and is denoted $|M|$.
- $\lambda = [\lambda_1, \lambda_2, \dots, \lambda_k]$ is a **partition** of n if $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$ and $\lambda_1 + \lambda_2 + \dots + \lambda_k = n$.



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- We associate to a hypermap M three partitions of size $|M|$.

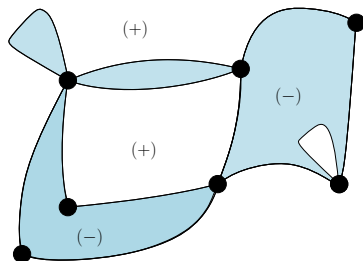


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The **vertex partition** of a hypermap M , denoted $\lambda^\bullet(M)$, is the partition obtained by dividing the degrees of the vertices by 2.

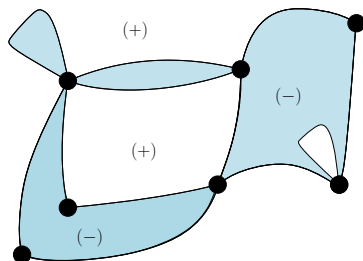


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The **partition of (+) faces** of M , denoted $\lambda^+(M)$, is the partition obtained from the degrees of (+)-faces of M .

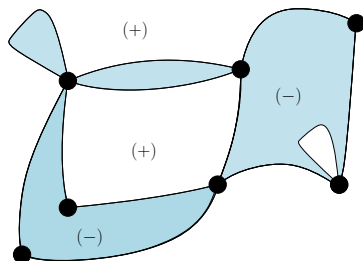


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The **partition of $(-)$ faces** of M , denoted $\lambda^-(M)$, is the partition obtained from the degrees of $(-)$ -faces of M .



$$|M| = 12, \quad \lambda^\bullet(M) = [3, 2, 2, 2, 1, 1, 1], \quad \lambda^+(M) = [7, 4, 1], \\ \lambda^-(M) = [5, 4, 2, 1].$$

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- We associate to a hypermap M three partitions of size $|M|$.
- The three partitions $(\lambda^\bullet(M), \lambda^+(M), \lambda^-(M))$ form the **profile** of M .

Weight of a hypermap

- We consider a variable t , and three alphabets of variables $\mathbf{p} := (p_1, p_2, \dots)$, $\mathbf{q} := (q_1, q_2, \dots)$ and $\mathbf{r} := (r_1, r_2, \dots)$. If $\lambda = [\lambda_1, \dots, \lambda_\ell]$ is a partition, then

$$p_\lambda := p_{\lambda_1} \cdots p_{\lambda_\ell}.$$

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 - ▶ an edge a weight t .
 - ▶ a vertex of degree $2i$ a weight p_i .
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- A hypermap M has then a weight $t^{|M|} p_{\lambda^\bullet(M)} q_{\lambda^+(M)} r_{\lambda^-(M)}$.

Generating series

$$H^{(1)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) = \sum_{\substack{\text{Orientable} \\ \text{hypermaps } M}} \frac{1}{z_{\lambda^\bullet(M)}} t^{|M|} p_{\lambda^\bullet(M)} q_{\lambda^+(M)} r_{\lambda^-(M)},$$

$$H^{(2)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) = \sum_{\substack{\text{Hypermaps } M \\ \text{orientable or not}}} \frac{1}{2^{\ell(\lambda^\bullet(M))} z_{\lambda^\bullet(M)}} t^{|M|} p_{\lambda^\bullet(M)} q_{\lambda^+(M)} r_{\lambda^-(M)}.$$

Remarque

In order to eliminate symmetries, we consider maps with some labelling.

Differential equations for some specializations of $H^{(i)}$

We consider the specialization $H^{(i)}(t, \mathbf{p}, \mathbf{q}, u)$ in which we keep two alphabets $\mathbf{p}$ and $\mathbf{q}$, and we replace $r_j = u$ for any $j \geq 1$. Equivalently, we associate to a hypermap M the weight

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- Chapuy-Dolega '22 : Differential equations for a one parameter deformation $H^{(\alpha)}$, related to the enumeration of maps with some non-orientability weights.

$$\frac{n\partial}{\partial q_n} \cdot H^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, u) = t^n \mathcal{B}_n^{(\alpha)}(\mathbf{p}, u) \cdot H^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, u).$$

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Question: Do the full series $H^{(i)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r})$ satisfy similar differential equations?

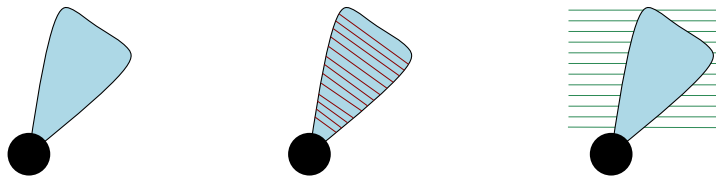
Hypermaps with some marked faces

We consider hypermaps with some marked faces of degree 1, with the condition that we do not simultaneously mark the two faces of an isolated loop.

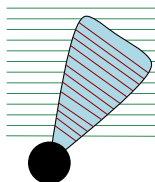
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Possible face marking of an isolated loop



Prohibited face marking of an isolated loop



→ Each edge is incident to at most one marked face.

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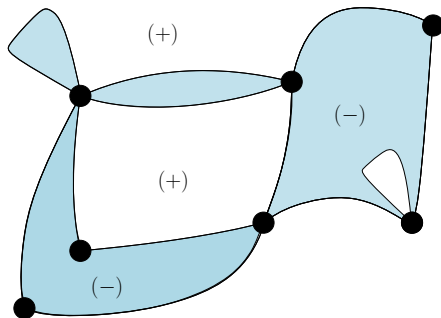


Figure 2: A hypermap of profile $([3, 2, 2, 2, 1, 1, 1], [7, 4, 1], [5, 4, 2, 1])$

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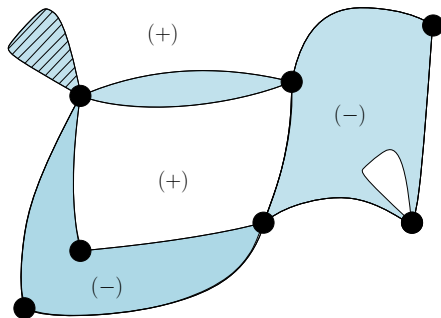


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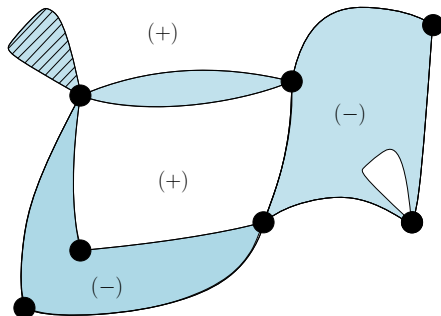


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The size of a hypermap with marked faces is

$|M| := |\lambda^+(M)| + |\lambda^-(M)| - |\lambda^\bullet(M)|$ is the number of edges of M which are not incident to a marked face.

Generating series

We associate to hypermaps with marked faces the generating series

$$\tilde{H}^{(1)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) := \sum_{\substack{\text{Orientable maps } M \\ \text{with marked faces}}} \frac{1}{z_{\lambda^\bullet(M)}} t^{|M|} p_{\lambda^\bullet(M)} q_{\lambda^+(M)} r_{\lambda^-(M)}.$$

It is related to the series of (ordinary) orientable hypermaps by

$$\exp\left(\frac{p_1}{t}\right) \tilde{H}^{(1)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) = \exp\left(\frac{\partial}{t\partial q_1}\right) \exp\left(\frac{\partial}{t\partial r_1}\right) H^{(1)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}),$$

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In a similar way, we define $\tilde{H}^{(2)}$ as the series of hypermaps with marked faces, orientable or not .

Main result:

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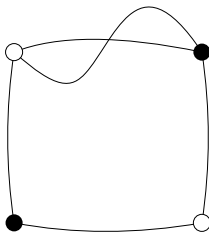
- ▶ For $\alpha = 1$, $G^{(1)} = \widetilde{H}^{(1)}$ is the series of orientable hypermaps with marked faces.
- ▶ For $\alpha = 2$, $G^{(2)} = \widetilde{H}^{(2)}$ is the series of all hypermaps (orientable or not) with marked faces.

$\mathcal{C}_\ell^{(\alpha)}$ is a differential operator.

Bipartite maps

A **bipartite** map is a map whose vertices are colored in black and white and such that each edge connects two vertices of different colors (obtained from hypermaps by duality).

Example: An orientable bipartite map



We associate to a bipartite map M of face partition $\lambda^\diamond(M)$, a weight $t^{|M|} p_{\lambda^\diamond(M)}$.

Differential operators and maps

Fix a bipartite map M .

$$\left(\sum_{i \geq 1} p_{i+1} \frac{i \partial}{\partial p_i} \right) p_{\lambda^\diamond(M)} = \sum_e p_{\lambda^\diamond(M \cup \{e\})}$$

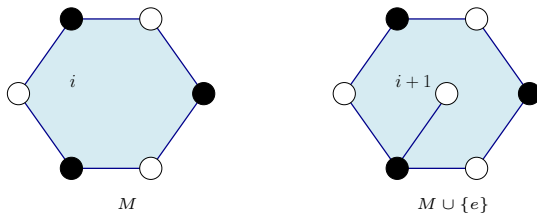
the sum is taken over all possible ways to add a white leaf to a black corner of M .

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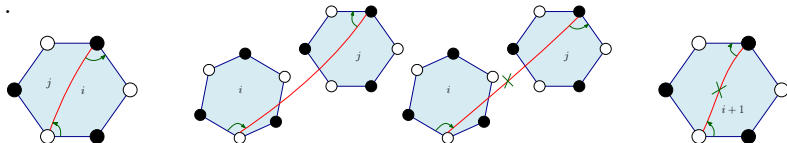
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Differential operators and maps

$$\left(\sum_{i,j \geq 1} p_i p_j \frac{(i+j-1)\partial}{\partial p_{i+j-1}} + 2 \sum_{i,j \geq 1} p_{i+j+1} \frac{i\partial}{\partial p_i} \frac{j\partial}{\partial p_j} + \sum_{i \geq 1} p_{i+1} \frac{i\partial}{\partial p_i} \right) \cdot p_{\lambda^\diamond(M)} = \sum_e p_{\lambda^\diamond(M \cup \{e\})}$$

the sum is taken over all possible ways to add an edge e between two corners of M .



Differential operators [Chapuy–Dołęga '22]

We fix a bipartite map M .

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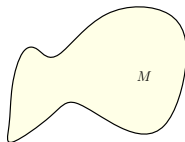
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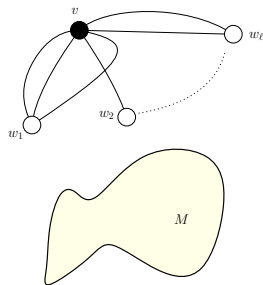


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$$\mathcal{C}_\ell^{(1)}(t, \mathbf{p}) \cdot \frac{t^{|M|} p_{\lambda^\diamond(M)}}{|M|!} = \sum_N t^{|N|} \frac{p_{\lambda^\diamond(N)}}{|N|!},$$

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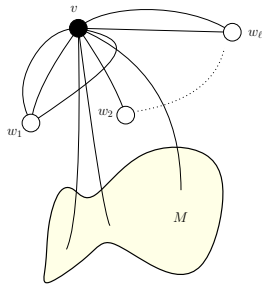


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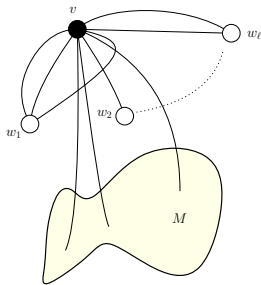


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→ These operators have a differential expression.

Differential operators [Chapuy–Dołęga '22]

$$\mathcal{C}_{\ell,k}^{(\alpha)}(\mathbf{p}) := [u^\ell] \Theta_Y (\Gamma_Y + uY_+)^{\ell+k} \frac{y_0}{1+b}$$

and

$$\mathcal{C}_\ell^{(\alpha)}(\mathbf{p}) := \sum_{k \geq 0} \frac{t^{\ell+k}}{\ell+k} \mathcal{C}_{\ell,k}^{(\alpha)}, \text{ for } \ell \geq 1.$$

where $b := \alpha - 1$,

$Y := (y_0, y_1, y_2, \dots)$ is a catalytic alphabet,

$$Y_+ = \sum_{i \geq 0} y_{i+1} \frac{\partial}{\partial y_i},$$

$$\Gamma_Y = (1+b) \cdot \sum_{i,j \geq 1} y_{i+j} \frac{i \partial^2}{\partial p_i \partial y_{j-1}} + \sum_{i,j \geq 1} y_i p_j \frac{\partial}{\partial y_{i+j-1}} + b \cdot \sum_{i \geq 1} y_{i+1} \frac{i \partial}{\partial y_i},$$

$$\Theta_Y := \sum_{i \geq 1} p_i \frac{\partial}{\partial y_i}.$$

Main result:

$$\begin{aligned} & \left(\mathcal{C}_\ell^{(\alpha)}(-t, \mathbf{q}) + \mathcal{C}_\ell^{(\alpha)}(-t, \mathbf{r}) \right) \cdot G^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) \\ &= \mathcal{C}_\ell^{(\alpha)\perp}(-t, \mathbf{p}) \cdot G^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}), \quad \text{for } \ell \geq 0; \end{aligned}$$

$G^{(\alpha)}$ is a deformation of the series of hypermaps with controlled degrees of vertices, faces (+) and faces (-), and marked faces of degree 1.

- ▶ For $\alpha = 1$, $G^{(1)} = \tilde{H}^{(1)}$ is the series of orientable hypermaps with marked faces.
- ▶ For $\alpha = 2$, $G^{(2)} = \tilde{H}^{(2)}$ is the series of all hypermaps (orientable or not) with marked faces.

$\mathcal{C}_\ell^{(\alpha)}$ is a differential operator.

Jack characters $\theta_{\mu}^{(\alpha)}(\lambda)$

- give the expansion of **Jack polynomials** $J_{\lambda}^{(\alpha)}$ in the power-sum basis,

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- form a basis of the space of shifted symmetric functions,
- count **layered maps** with non-orientability weights.

Theorem (B.D–Dołęga '23+)

For $\lambda = [\lambda_1, \dots, \lambda_k]$

$$\theta_\mu^{(\alpha)}(\lambda) = [t^{|\mu|} p_\mu] \exp \left(\mathcal{C}_\infty^{(\alpha)}(-t, \mathbf{p}, -\alpha \lambda_1) \right) \cdots \exp \left(\mathcal{C}_\infty^{(\alpha)}(-t, \mathbf{p}, -\alpha \lambda_k) \right) \cdot 1,$$

where $\mathcal{C}_\infty^{(\alpha)}(t, \mathbf{p}, u) = \sum_{\ell \geq 0} u^\ell \mathcal{C}_\ell^{(\alpha)}(t, \mathbf{p})$.

For $\alpha = 1$, this corresponds to Stanley–Féray formula '10 for the characters of the symmetric group.

Structure coefficients

$$\theta_{\mu}^{(\alpha)} \cdot \theta_{\nu}^{(\alpha)} = \sum_{|\pi| \leq |\mu| + |\nu|} g_{\mu, \nu}^{\pi}(\alpha) \theta_{\pi}^{(\alpha)}.$$

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Theorem

$g_{\mu, \nu}^{\pi}(1)$ is the number of *orientable* hypermaps with marked faces and with profile (π, μ, ν) .

$g_{\mu, \nu}^{\pi}(2)$ is the number of hypermaps *orientable or not*, with marked faces and with profile (π, μ, ν) .

Goulden–Jackson '96 + Dołęga–Féray '16 + Definition of hypermaps with marked faces.

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We introduce the series

$$G^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) := \sum_{\pi, \mu, \nu} t^{|\mu| + |\nu| - |\pi|} \frac{g_{\mu, \nu}^{\pi}(\alpha)}{z_{\pi} \alpha^{\ell(\pi)}} p_{\pi} q_{\mu} r_{\nu}.$$

$$\longrightarrow G^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) = \tilde{H}^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) \quad \text{for } \alpha \in \{1, 2\}.$$

Theorem (B.D. '24)

$$\begin{aligned} \left(\mathcal{C}_\ell^{(\alpha)}(-t, \mathbf{q}) + \mathcal{C}_\ell^{(\alpha)}(-t, \mathbf{r}) \right) \cdot G^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) \\ = \mathcal{C}_\ell^{(\alpha)\perp}(-t, \mathbf{p}) \cdot G^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}), \quad \text{for } \ell \geq 0. \end{aligned}$$

where $\mathcal{C}_\ell^{(\alpha)\perp}$ is the dual of $\mathcal{C}_\ell^{(\alpha)}$ with respect to the scalar product

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Proof : We use

$$\theta_\mu^{(\alpha)} \cdot \theta_\nu^{(\alpha)} = \sum_{\pi} g_{\mu, \nu}^{\pi}(\alpha) \theta_\pi^{(\alpha)},$$

$$\theta_\mu^{(\alpha)}(\lambda) = [t^{|\mu|} p_\mu] \exp \left(\mathcal{C}_\infty^{(\alpha)}(-t, \mathbf{p}, -\alpha \lambda_1) \right) \cdots \exp \left(\mathcal{C}_\infty^{(\alpha)}(-t, \mathbf{p}, -\alpha \lambda_k) \right) \cdot 1,$$

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Commutation relations

$$G^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) := \sum_{\pi, \mu, \nu} t^{|\mu|+|\nu|-|\pi|} \frac{g_{\mu, \nu}^{\pi}(\alpha)}{z_{\pi} \alpha^{\ell(\pi)}} p_{\pi} q_{\mu} r_{\nu}.$$

We introduce the operator

$$\begin{aligned} \mathcal{G}^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) : \quad \mathbb{Q}(\alpha)[\mathbf{p}] &\longrightarrow \mathbb{Q}(\alpha)[\mathbf{q}, \mathbf{r}][[t]] \\ p_{\pi} &\longmapsto \sum_{\mu, \nu} t^{|\mu|+|\nu|-|\pi|} g_{\mu, \nu}^{\pi}(\alpha) q_{\mu} r_{\nu}. \end{aligned}$$

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The differential equations are equivalent to the following commutation relations

$$\left(\mathcal{C}_{\ell}^{(\alpha)}(-t, \mathbf{q}) + \mathcal{C}_{\ell}^{(\alpha)}(-t, \mathbf{r}) \right) \cdot \mathcal{G}^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) = \mathcal{G}^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) \cdot \mathcal{C}_{\ell}^{(\alpha)}(-t, \mathbf{p}),$$

for $\ell \geq 0$.

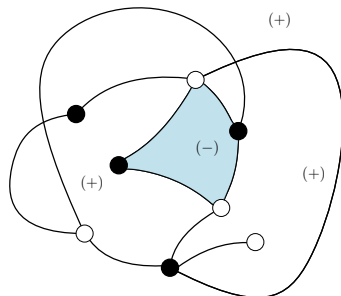
Combinatorial proof for $\alpha = 1$

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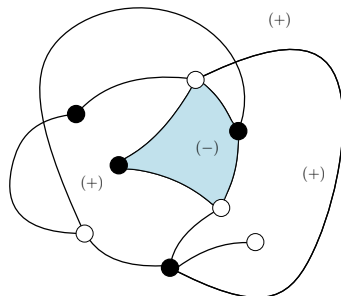
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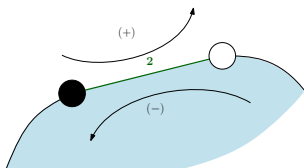
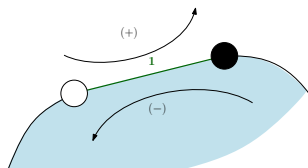
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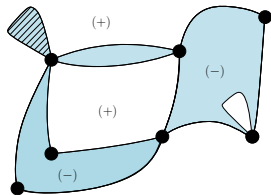
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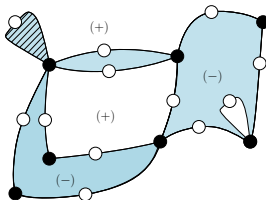
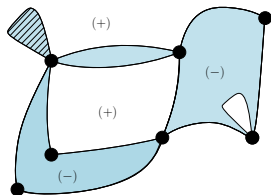
We distinguish two types of bicolor edges:



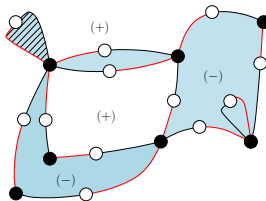
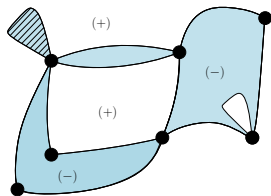
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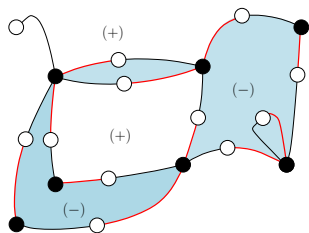
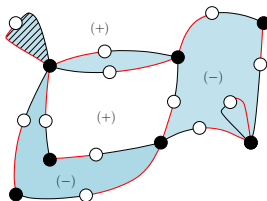
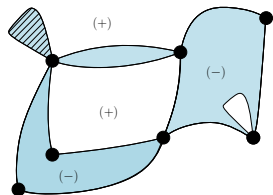
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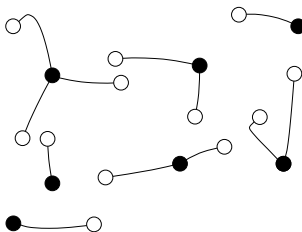
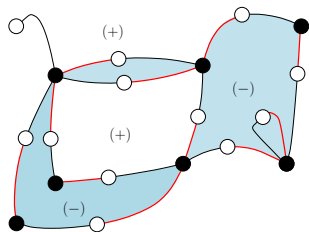
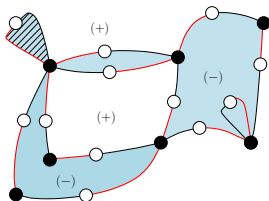
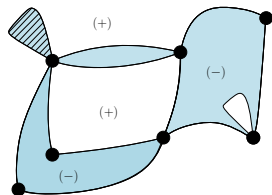
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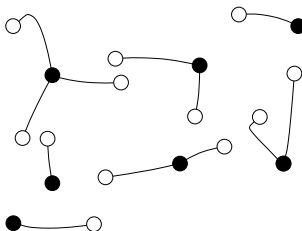
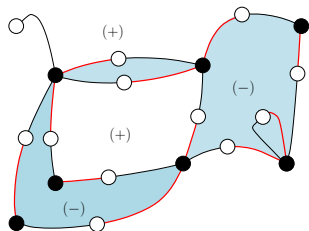
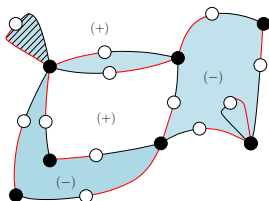
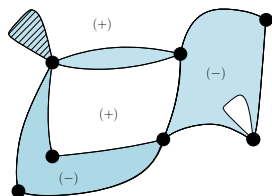
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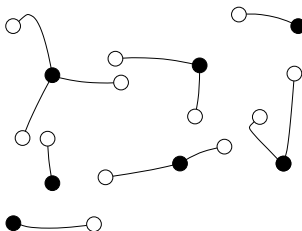
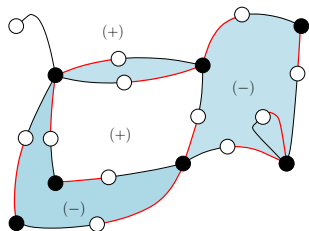
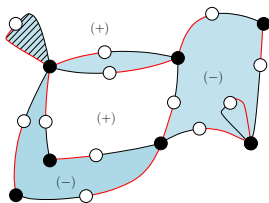
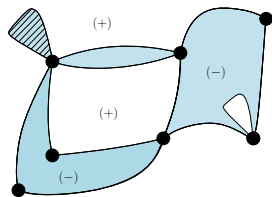
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$g_{\mu,\nu}^{\pi}(1)$ counts the number of ways of:

- adding edges to π -star map.
- coloring faces s.t. all added edges are bicolor and of type 2,
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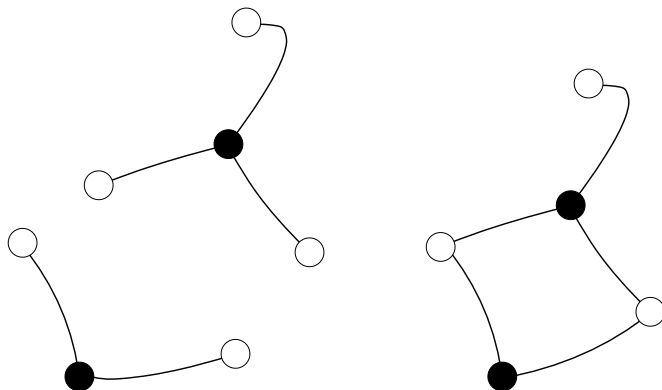
$g_{\mu, \nu}^{\pi}(1)$ counts the number of ways of:

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Combinatorial proof for $\alpha = 1$

There are the same number of ways of adding edges and color faces on two bipartite maps of the same face partition.

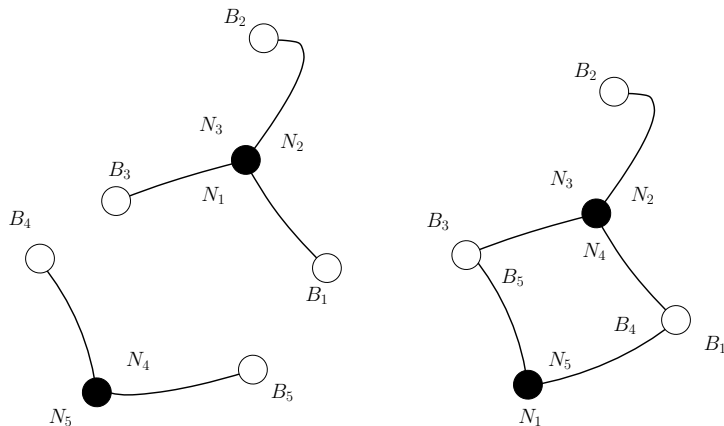
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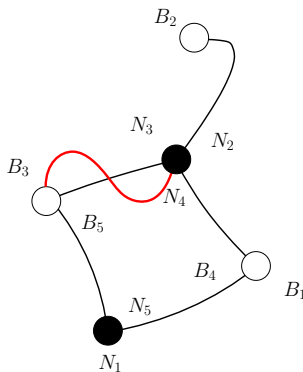
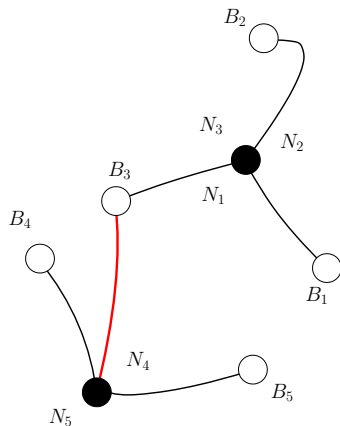
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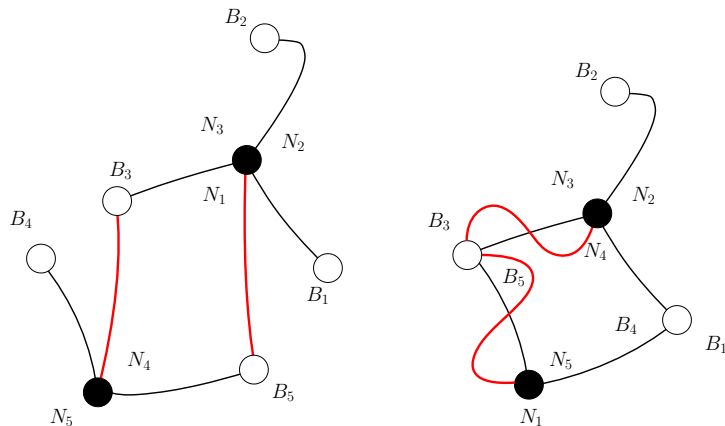
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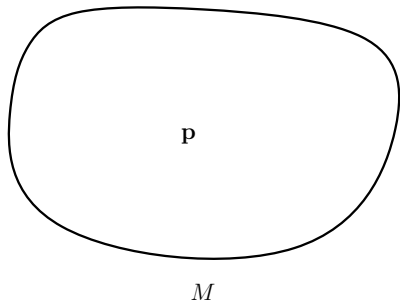
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We fix a bipartite map M .

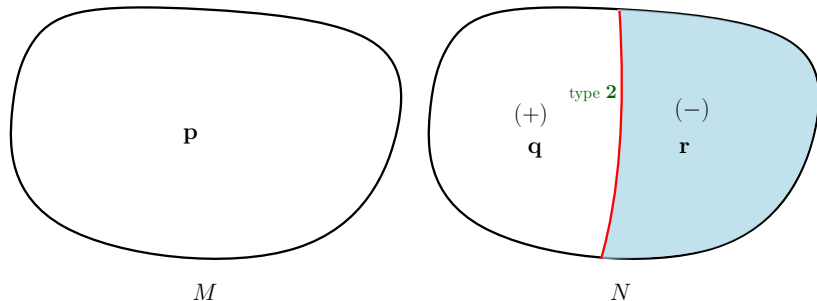
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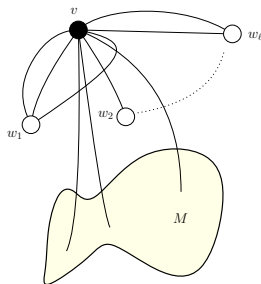
$$\left(\mathcal{C}_\ell^{(1)}(-t, \mathbf{q}) + \mathcal{C}_\ell^{(1)}(-t, \mathbf{r})\right) \cdot \mathcal{G}^{(1)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) = \mathcal{G}^{(1)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) \cdot \mathcal{C}_\ell^{(1)}(-t, \mathbf{p}).$$

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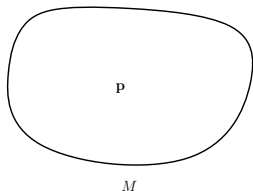
Recall:

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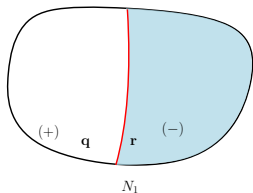
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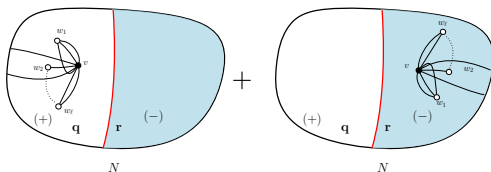
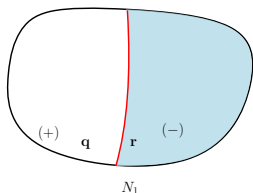
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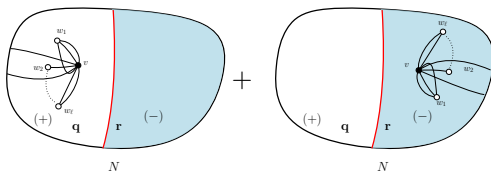
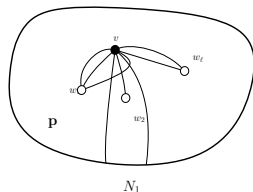
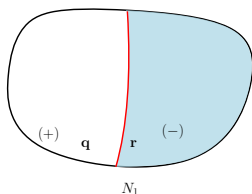
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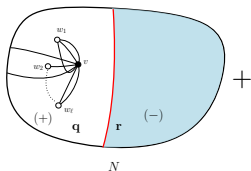
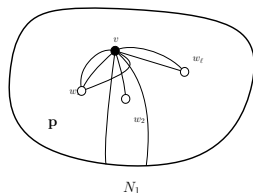
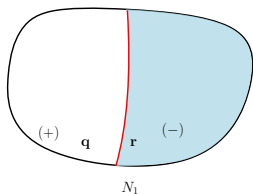
Combinatorial proof for $\alpha = 1$

$$\left(\mathcal{C}_\ell^{(1)}(-t, \mathbf{q}) + \mathcal{C}_\ell^{(1)}(-t, \mathbf{r})\right) \cdot \mathcal{G}^{(1)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) = \mathcal{G}^{(1)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) \cdot \mathcal{C}_\ell^{(1)}(-t, \mathbf{p}).$$

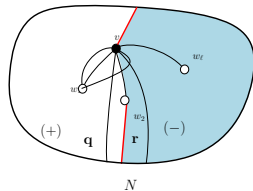
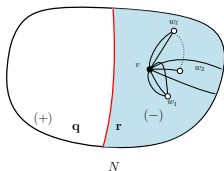


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+



Positivity conjecture

Conjecture (Goulden–Jackson'96+Śniady '19)

*The coefficients $g_{\mu,\nu}^{\pi}$ are **polynomials** in $b := \alpha - 1$ with **non-negative** and **integer** coefficients. Moreover, $g_{\mu,\nu}^{\pi}(\alpha)$ counts hypermaps with non-orientability weights.*

The Matching-Jack conjecture of Goulden–Jackson corresponds to the case $|\pi| = |\mu| = |\nu|$.

- Polynomiality : Dołęga–Féray '16.
- Integrality : B.D. '22.
- Positivity ??

Resolution of the differential equation

Theorem (B.D. '24)

$$g_{\mu,\nu}^{\lambda}(\alpha) = (-1)^{|\mu|+|\nu|-\lambda} \sum_{m \geq 0} (-1)^m \sum_{\substack{\pi_1, \dots, \pi_m \\ |\pi_1| < \dots < |\pi_m|}} a_{\pi_1}^{\lambda} a_{\pi_2}^{\pi_1} a_{\pi_3}^{\pi_2} \dots a_{\pi_m}^{\pi_{m-1}} d_{\mu,\nu}^{\pi_m},$$

where

$$a_{\xi}^{\lambda} := [t^{|\xi|} p_{\xi}] \left(\prod_{i=1}^{\ell(\lambda)} \alpha \lambda_i \mathcal{C}_{\lambda_i}^{(\alpha)}(t, \mathbf{p}) \right) \cdot 1$$

and

$$d_{\mu,\nu}^{\lambda} := [t^{|\mu|+|\nu|} q_{\mu} r_{\nu}] \prod_{1 \leq i \leq \ell(\mu)} \alpha \lambda_i \left(\mathcal{C}_{\lambda_i}^{(\alpha)}(t, \mathbf{q}) + \mathcal{C}_{\lambda_i}^{(\alpha)}(t, \mathbf{r}) \right) \cdot 1,$$

Applications

$$\mathcal{G}^{(\alpha)}(t, \mathbf{p}, \mathbf{q}, \mathbf{r}) = \sum_{k \geq 0} t^k \mathcal{G}_k^{(\alpha)}(\mathbf{p}, \mathbf{q}, \mathbf{r}).$$

Theorem (B.D. '24)

$$\mathcal{G}_0^{(\alpha)} = \Psi, \quad \text{où } \Psi \cdot p_\lambda = \prod_{1 \leq i \leq \ell(\lambda)} (q_i + r_i).$$

$$\mathcal{G}_1^{(\alpha)} = \sum_{m \geq 1} \sum_{\substack{m_1 + m_2 = m+1 \\ m_1, m_2 \geq 1}} q_{m_1} r_{m_2} \cdot \Psi \cdot \frac{m \partial}{\partial p_m},$$

$$\begin{aligned} \mathcal{G}_2^{(\alpha)} = & \frac{1}{2} \sum_{m \geq 1} \sum_{\substack{m_1 + m_2 = m+2 \\ m_1, m_2 \geq 1}} b(m_1 - 1)(m_2 - 1) q_{m_1} r_{m_2} \Psi \frac{m \partial}{\partial p_m} \\ & + \frac{1}{2} \sum_{m \geq 1} \sum_{\substack{m_1 + m_2 + m_3 = m+2 \\ m_1, m_2, m_3 \geq 1}} (m_1 - 1)(q_{m_1} r_{m_2} r_{m_3} + r_{m_1} q_{m_2} q_{m_3}) \Psi \frac{m \partial}{\partial p_m} \\ & + \frac{1}{2} \sum_{k, m \geq 1} \sum_{\substack{i_1 + i_2 = k+m+2 \\ i_1, i_2 \geq 1}} \alpha \min(m, k, i_1 - 1, i_2 - 1) q_{i_1} r_{i_2} \Psi \frac{m \partial}{\partial p_m} \frac{k \partial}{\partial p_k} \\ & + \frac{1}{2} \sum_{m, k \geq 1} \sum_{\substack{m_1 + m_2 = m+1 \\ m_1, m_2 \geq 1}} \sum_{\substack{k_1 + k_2 = k+1 \\ k_1, k_2 \geq 1}} q_{m_1} q_{k_1} r_{m_2} r_{k_2} \Psi \frac{m \partial}{\partial p_m} \frac{k \partial}{\partial p_k}. \end{aligned}$$